Koneru Lakshmaiah Education Foundation
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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-7	Team No-8	Project Title - Library Mobile App for member Self Service

1. Abstract

Library Mobile App for Member Self-Service is a database-based application designed to provide library members with convenient access to library services through a mobile application. The system will allow members to search for books, check book availability, view their borrowing history, reserve books, and monitor due dates and fines without depending on library staff.

The proposed system will maintain details of members, books, authors, categories, borrowings, reservations, returns, and fines in a structured relational database. The application will allow users to search and filter books, check availability, reserve available books, and view their current and previous transactions. Database operations such as SQL queries, joins, aggregation, constraints, and transactions will be used to maintain data consistency and generate useful library reports. The system will provide a simple and efficient self-service platform that improves the overall library experience for members.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520030268	A RISHI	
2520030093	K SRI CHARAN	
2520030455	P SUDHA	

3. Frameworks / Platforms / Software Used

PostgreSQL – For storing and managing members, books, borrowing, reservations, returns, and fines.
SQL – For database creation, queries, joins, aggregation, constraints, and transactions.
FastAPI – Backend framework for developing REST APIs and connecting the application with the database.
Python – Backend programming language.
SQLAlchemy – For database integration and interaction with PostgreSQL.
Postman – For testing and validating REST APIs.

Guide Name and Signature (remarks if any) _____